

Course Title: Database Management Systems

Course Code: CSE-2423

Course Content of Semester Final Exam

Group-A (20 Marks)

Segment 4. Integrity, Security and Relational Database Design:

Domain constraint, Integrity, Assertions, Triggers, Authorization, Authentication, Security, Privileges, Roles, and Audit trails, Encryption-Decryption Algorithm, Decomposition etc.

Segment 5. Functional Dependency and Normalization:

Functional Dependencies, Closure of a set of Functional dependencies. Un-normal Form (UNF), First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce and Codd Normal Form (BCNF).

Group-B (30 Marks)

Segment 6. Indexing and Hashing:

Ordered indices, Hash indices, Hash function, Primary index, Secondary index, Dense, sparse, Multilevel indices, B+ tree index files, Handling Bucket Overflows, Overflow Chaining, Closed Hashing, Open Hashing, Linear probing, Hash indices, Dynamic Hashing.

Segment 7. Transaction:

ACID Properties, Transaction state diagram, Implementation of Atomicity and Durability, Shadow copy technique, Concurrent Execution, Serializability, Recoverability, Recoverable schedule, Cascadeless Schedules, Implementation in Isolation, Testing of Serializability.

Segment 8. Concurrency control, Recovery System and Distribute databases:

Lock-Based Protocols, Granting of locks, Two-phase locking protocol, Graph based protocol, Tree protocol, Timestamp based protocols, Deadlock detection and recovery. Failure classification, Storage types, Checkpoints. Distributed data, Replication and Fragmentation.

 Combined Question Analysis (Sp24 + Au24 + Sp25)

Question Focus	Sp24 Questions	Au24 Questions	Sp25 Questions	Syllabus Segment	Topic & Concept
SQL DDL, Constraints, Triggers	Q1(a), Q1(b)	Q1(a), Q1(b)	Q1(a), Q1(b)	Segment 4: Integrity, Security & Authorization	DDL schema creation with NOT NULL, CHECK, UNIQUE, FOREIGN KEY; Trigger/Assertion ensures data integrity — appears every term.
Domain & Integrity Constraints	Q1(a)	Q1(a)	Q1(a)	Segment 4	Domain constraint, referential integrity, key-based constraint — consistent in all semesters.
Encryption, Authorization, Roles	—	Q1(a)	Q1(b) (Auth vs Authen.)	Segment 4	Appears occasionally; Autumn 2024 emphasized roles and privileges; Spring 2025 revisited authorization.
Normalization (1NF–BCNF)	Q1(a, alt), Q2(a)	Q2(a), Q2(b)	Q1(a), Q2(a)	Segment 5: Functional Dependency & Normalization	Core area — normalization to 3NF/BCNF with justification, key identification, and decomposition.
Functional Dependency, Closure, Candidate Keys	Q2(a)	Q2(a)	Q2(a,b)	Segment 5	FD set analysis, closure computation, key checking — repeatedly tested; often carries 6–8 marks.
Dependency Preservation & Lossless Join	Q2(b)	Q2(b)	Q1(a, 2NF part)	Segment 5	Theory-based — decomposition property and equivalence; at least one short question every term.
Dense vs Sparse Index, Hash Function	—	Q3(a)	Q3(a)	Segment 6: Indexing & Hashing	Conceptual index comparison — dense/sparse or

					hash-based structure analysis.
B+ Tree Construction & Deletion	—	Q3(b)	Q3(b)	Segment 6	Step-by-step B+ tree insertion/deletion — common in both 2024 and 2025; carries 8–10 marks.
ACID Properties & Transaction States	Q4	Q4(b)	Q4(a,b)	Segment 7: Transactions	Fundamental ACID explanation and transaction lifecycle/state diagram; always tested.
Serial & Serializable Schedule, Precedence Graph	Q4	Q4(a)	Q4(c)	Segment 7	Conflict serializability with precedence graph — identical question trend across semesters.
Cascadeless Schedule & Recoverability	Q4	—	Q4(iv)	Segment 7	Schedule property analysis — occasionally tested with ACID.
Concurrency Control (2PL, Deadlocks, Timestamping)	—	Q5(a,b,c)	Q5(a,b)	Segment 8: Concurrency Control	Locking and deadlock schemes dominate; timestamp ordering reappears in Spring 2025.
Database Recovery (Logging, Checkpoint, Failure Recovery)	Q5(b,c)	—	Q5(c)	Segment 8: Recovery System	Recovery techniques (log-based, checkpoint) — recurring last question theme.
Distributed Database (Replication & Fragmentation)	Q5(a)	—	—	Segment 8: Distributed Databases	Appeared only in Spring 2024 — less frequent but still within syllabus.

Topic Weight Comparison (Across Three Semesters)

Syllabus Segment	Topic Coverage	Sp24 Focus	Au24 Focus	Sp25 Focus
Segment 4: Integrity, Security, Authorization	SQL, Constraints, Triggers, Security	☑ Strong	☑ Strong	☑ Strong
Segment 5: Functional Dependency & Normalization	FD, 1NF–BCNF, Lossless, Closure	☑ Strong	☑ Very High	☑ Very High
Segment 6: Indexing & Hashing	Dense/Sparse Index, B ⁺ Tree	○ Absent	☑ Strong	☑ Very Strong
Segment 7: Transactions & Serializability	ACID, States, Conflict Serializability	☑ Moderate	☑ Strong	☑ Strong
Segment 8: Concurrency, Recovery, Distributed DB	Locking, Timestamp, Logging, Checkpoint	☑ Moderate	☑ Moderate	☑ Strong

Insights

1. **Normalization + FD** → Consistent 10 marks every term; BCNF, closure, and dependency questions are guaranteed.
2. **SQL + Constraints + Triggers** → Appears in Q1 every semester; practical schema design focus.
3. **B⁺ Tree Indexing** → Always in Group B since Autumn 2024; step-by-step insertion/deletion format.
4. **Transactions** → ACID, conflict serializability, and state diagrams are fixed pattern questions.
5. **Concurrency + Recovery** → Final question in all terms, alternating between 2PL/deadlock and checkpoint/recovery.